

Early Warning Student Dropout Risk Prediction System

Pillar 5: Capstone Project - Postgraduate Diploma in Artificial Intelligence and Machine Learning

Item	Detail
Author	Rex Villamar
Date	October 2026
Domain	Education: predicting student dropout (supervised binary classification)
Dataset	UCI Machine Learning Repository, Predict Students' Dropout and Academic Success (Realinho et al., 2022); 4,424 students, 36 features
Code repository	https://github.com/theboneidentity/PILLAR_5_CAPSTONE_PROJECT

Abstract

Student dropout costs institutions tuition revenue and costs students their qualification. This report documents an end-to-end machine learning project that scores each student at the end of Semester 2 and flags those likely to drop out, so advisors can intervene while later semesters remain. Using 4,424 student records, four model families were tuned with stratified cross-validation and ranked on F2, a metric that weights recall twice as heavily as precision because a missed dropout is far costlier than an unnecessary advisor check-in. The selected XGBoost model reaches recall 0.912, precision 0.685, F2 0.855, ROC-AUC 0.937 and PR-AUC 0.911 on a held-out set of 885 students, meeting the recall target of 0.85. The strongest drivers are the Semester 2 and Semester 1 approval rates, followed by tuition status, age and course. A five-attribute fairness audit found recall gaps for age group (0.18) and scholarship holders (0.13); age-specific decision thresholds reduced the age gap to 0.074 at the cost of lower precision and a wider false-positive gap, and the scholarship gap remains open. All results come from a single reproducible notebook, and the limitations and open issues are reported alongside the results.

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1. Problem Understanding and Framing

Higher-education institutions lose tuition revenue, and students lose their qualification, when students withdraw unexpectedly. Acquiring a new student typically costs more than retaining an existing one, so detecting dropout risk early and delivering targeted support, such as tutoring, counselling or financial-aid review, has clear value. The question this project answers is: *at the end of Semester 2, which students are most likely to drop out?*

1.1 Task definition

- **Task type:** supervised binary classification. The target is $y = 1$ for Dropout and $y = 0$ for Retained (Graduate or Enrolled).

- **Decision point:** the end of Semester 2. Each student is scored once their first- and second-semester records are complete, which leaves later semesters in which an intervention can still change the outcome. Every input feature is available by that point.
- **Responsible-AI constraint:** Gender and Marital status are not model inputs; they are kept only to audit whether outcomes differ across groups.

1.2 Cost asymmetry and success metrics

Actual / predicted	Flagged as high risk	Not flagged
Dropout (y = 1)	True positive: intervention is possible	False negative: lost student (high cost)
Retained (y = 0)	False positive: advisory check-in (low cost)	True negative: no action needed

Because a missed dropout is far costlier than a false alarm, **recall on the dropout class** is the primary technical metric, with a deployment criterion of at least 0.85. The F2 score, $F2 = 5PR / (4P + R)$, ranks models; ROC-AUC and PR-AUC are reported as secondary metrics; precision is tracked as the price paid for recall. The business KPIs are the increase in student retention and the net tuition revenue saved after intervention cost (Section 8).

1.3 Evaluation protocol

A stratified 80/20 train/test split (`random_state = 42`) is made first. Models are tuned and selected on 5-fold stratified cross-validation of the training set only, preprocessing is fitted on training data only, and the held-out test set is scored once, at a fixed decision threshold of 0.50. This keeps the reported results free of test-set leakage.

2. Data Collection and Understanding

The data is the public UCI dataset *Predict Students' Dropout and Academic Success* (Realinho et al., 2022), which links enrolment records with first- and second-semester academic results for 4,424 students at one Portuguese higher-education institution. It was chosen because it is real, documented and public, and its columns match the decision to be supported. The notebook downloads it automatically.

- **Quality:** 0 missing values and 0 duplicate rows, so no imputation or de-duplication was required.
- **Target balance:** Graduate 49.9%, Dropout 32.1%, Enrolled 17.9%. Merging Graduate and Enrolled into Retained gives a binary problem with a 32.1% positive class, a moderate imbalance that must be handled in modelling and metric choice.
- **Encoding:** categorical variables (course, application mode, qualifications, occupations, nationality) arrive integer-coded; the UCI documentation gives the code meanings and the data dictionary in the notebook reports each column's type and range or cardinality.

Table 1. Feature groups in the data dictionary (36 features plus the target)

Group	Count	Examples
Academic (Semester 1 and 2)	12	curricular units credited, enrolled, evaluated, approved, grade, without evaluations (per semester)
Application and macroeconomic	10	application mode and order, course, attendance mode, previous qualification and grade, admission grade, unemployment, inflation, GDP
Demographic	7	marital status, nationality, gender, age at enrolment, international, displaced, special educational needs
Socio-economic	7	parents' qualifications and occupations, debtor, scholarship holder, tuition fees up to date

3. EDA and Feature Engineering Report

3.1 Cleaning audit

Nulls and duplicates were counted (none found). Outliers were screened with the 1.5 x IQR rule on grades, ages, unit counts and macroeconomic variables. Nine of twelve columns contain flagged values, up to 19.8% of rows. These were **retained**: they are genuine observations (for example mature students or many credited units), tree ensembles are insensitive to them, and the Logistic Regression model standardises its inputs inside its pipeline.

Table 2. Outlier screen (1.5 x IQR rule): values are flagged, not removed

Feature	Rows flagged	% of rows
Curricular units 2nd sem (grade)	877	19.82
Curricular units 1st sem (grade)	726	16.41
Age at enrollment	441	9.97
Curricular units 1st sem (enrolled)	424	9.58
Curricular units 2nd sem (enrolled)	369	8.34
Curricular units 1st sem (approved)	180	4.07
Previous qualification (grade)	179	4.05
Admission grade	86	1.94
Curricular units 2nd sem (approved)	44	0.99
Unemployment rate, Inflation rate, GDP	0	0.00

3.2 Domain-derived features

- **Approval rate** for each semester s: $\text{Approval_Rate_s} = \text{approved units} / (\text{enrolled units} + 0.00001)$, the share of the workload a student actually passes.
- **Academic momentum**: $\text{Grade_Trend} = \text{Grade (Semester 2)} - \text{Grade (Semester 1)}$.
- **Financial stress**: $\text{Financial_Stress_Index} = \text{Debtor} + (1 - \text{Tuition fees up to date}) + (1 - \text{Scholarship holder})$, an integer from 0 to 3.

All four features are computed row by row, so they cannot leak information across the train/test split.

3.3 Applied exploratory analysis

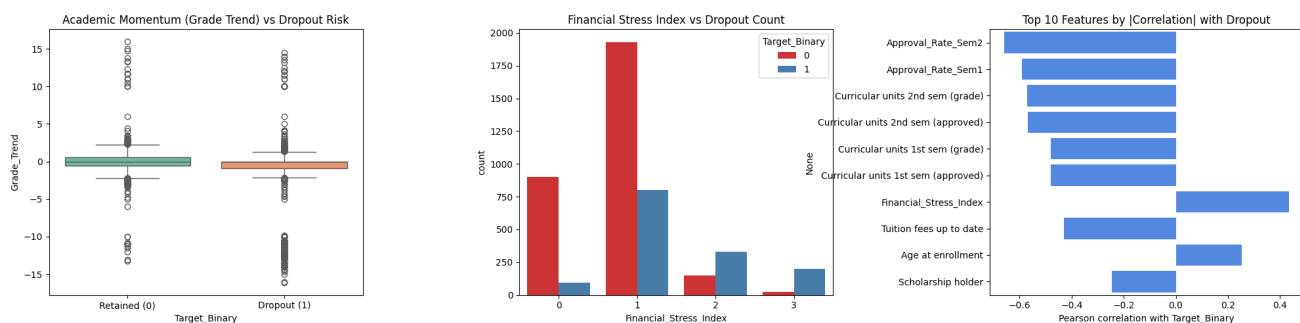


Figure 1. Grade_Trend by outcome (left), student counts by Financial_Stress_Index (centre) and the ten features most correlated with dropout (right).

- **Financial stress is steep**: the dropout rate is 9.3% at index 0, 29.3% at 1, 69.2% at 2 and 88.8% at 3.
- **Academic progress dominates**: the features most correlated with dropout are the Semester 2 and Semester 1 approval rates, then Semester 2 and 1 grades and approved units (all negative), followed by

Financial_Stress_Index, tuition status, age and scholarship status.

- **Academic momentum adds little:** the median Grade_Trend is 0.00 for both groups. The dropout distribution sits slightly lower, but the feature does not reach the top ten model drivers (Section 6).

3.4 Splitting, feature selection and dimensionality reduction

The stratified split gives 3,539 training and 885 test students, with a dropout rate of 32.1% in both. The scaler, RFECV and PCA are fitted on training data only. RFECV (a 50-tree random forest, 3-fold, scored on recall) retained all 38 features, so no feature was removed. PCA needs 24 components to retain 95% of the variance; the first two components explain 19.5% and 10.7%, and dropouts and retained students overlap heavily in that projection, indicating no clean linear separation (Figure 2). PCA is therefore used as a diagnostic and visualisation only: the models keep the original features so that SHAP explanations stay interpretable.

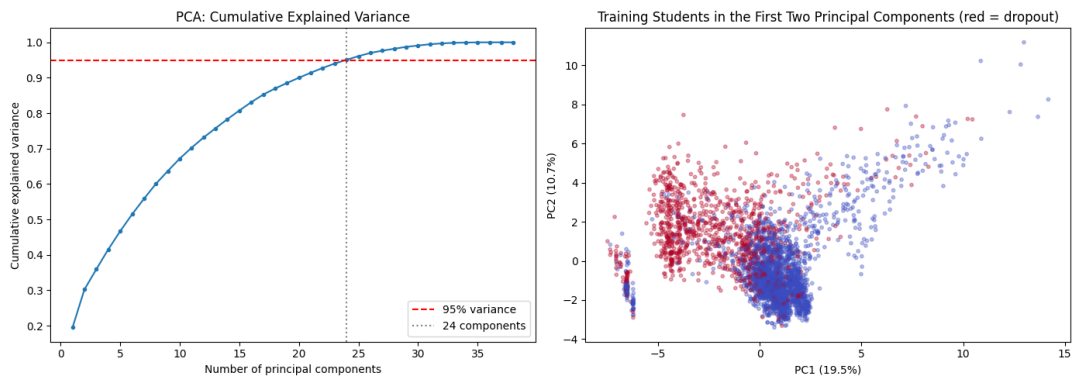


Figure 2. PCA diagnostics: cumulative explained variance (left) and training students in the first two components, red = dropout (right).

4. Model Implementation and Comparison

Four model families of increasing capacity were implemented. Logistic Regression is the linear baseline, with a StandardScaler inside its pipeline so scaling is re-fitted on every cross-validation training fold. Random Forest uses bagged decision trees. XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017) are gradient-boosted tree ensembles that can capture non-linear interactions. With only 32% dropouts, every grid included class weighting (class_weight for scikit-learn and LightGBM models, scale_pos_weight for XGBoost) so the minority class counts more in the loss.

Table 3. Hyperparameter search spaces (GridSearchCV, stratified 5-fold, scoring = F2)

Model	Parameters searched
Logistic Regression	C in {0.01, 0.1, 1, 10}; class_weight in {balanced, none}; solver in {liblinear, lbfgs}
Random Forest	n_estimators in {100, 200}; max_depth in {5, 10, none}; class_weight in {balanced, balanced_subsample, none}; min_samples_split in {2, 5}
XGBoost	n_estimators in {100, 200}; max_depth in {3, 5, 7}; learning_rate in {0.01, 0.1, 0.2}; scale_pos_weight in {1, 3, 5}
LightGBM	n_estimators in {100, 200}; max_depth in {3, 5, 7}; learning_rate in {0.01, 0.1, 0.2}; class_weight in {balanced, none}

Table 4. Model comparison, ranked by cross-validated F2 (hold-out metrics at threshold 0.50)

Model	CV F2 (mean +/- std)	Recall	Precision	F2	ROC-AUC	PR-AUC
XGBoost (selected)	0.825 +/- 0.020	0.912	0.685	0.855	0.937	0.911
Logistic Regression	0.806 +/- 0.023	0.835	0.780	0.823	0.928	0.898
LightGBM	0.803 +/- 0.015	0.852	0.783	0.837	0.935	0.910
Random Forest	0.793 +/- 0.021	0.835	0.785	0.824	0.929	0.897

Selection. Models were ranked on training-set cross-validated F2 only, so the test set played no part in the choice. XGBoost ranked first (best parameters: learning rate 0.1, maximum depth 3, 100 trees, scale_pos_weight 5) and also had the highest hold-out recall (0.912), meeting the 0.85 target. Its margin over the runner-up, Logistic Regression, is 0.019, smaller than one fold standard deviation (0.020), so the ranking is not statistically decisive and the choice is a judgement call. The other models trade six to eight points of recall for about ten points of higher precision. Every tuned model, the comparison table and a configuration file recording the parameters, threshold, seed, split and feature list are saved for reproducibility.

5. Evaluation and Limitations Audit

The selected model was audited on the same split and threshold. On the 885 hold-out students (284 actual dropouts) it catches 259 dropouts, misses 25, and raises 119 unnecessary check-ins, so it flags 378 students (42.7%).

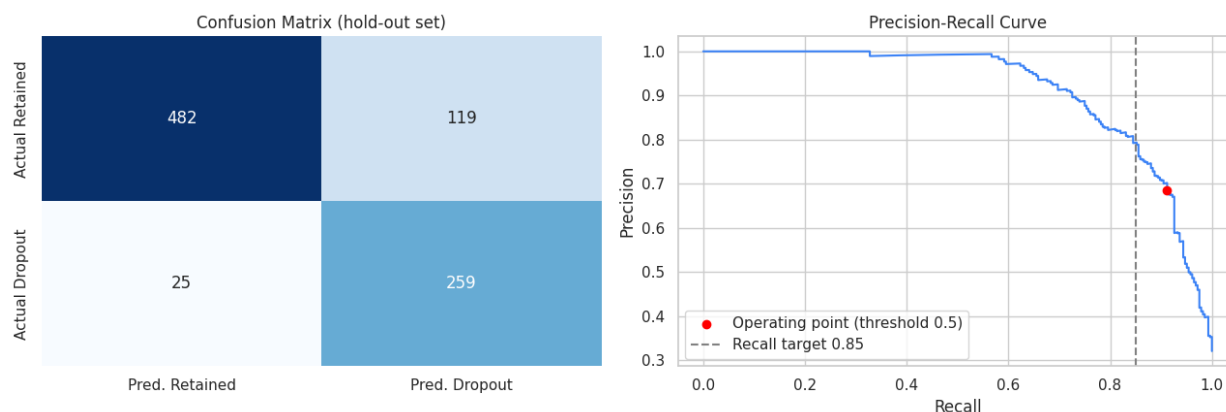


Figure 3. Confusion matrix (left) and precision-recall curve with the operating point and the 0.85 recall target (right), hold-out set.

5.1 Overfitting, leakage and timing

Table 5. Overfitting check: stratified 5-fold cross-validation on the training set

Metric	Train	Validation	Gap
Recall	0.932	0.880	+0.053
Precision	0.699	0.661	+0.038
F2	0.874	0.825	+0.049
ROC-AUC	0.959	0.923	+0.036

- **Generalisation:** the train-minus-validation F2 gap is 0.049, below the 0.10 level treated as overfitting, so the generalisation gap is acceptable.

- **Leakage screen:** no single feature exceeds 0.90 ROC-AUC on its own; the strongest is Approval_Rate_Sem2 at 0.871, followed by units approved in Semester 2 (0.853), Approval_Rate_Sem1 (0.836), units approved in Semester 1 (0.816) and Semester 2 grade (0.805).
- **Feature timing:** of the 38 inputs, 23 are known at enrolment, 7 at the end of Semester 1 and 8 at the end of Semester 2, so none uses information from the future.
- **Value of Semester 2 data:** the same model trained without Semester 2 features reaches ROC-AUC 0.912 and recall 0.880; with them, 0.937 and 0.912.
- **Class imbalance:** dropouts are 32.1% of training rows; class weighting raises recall and its cost is the precision reported above.

6. Explainability

SHAP (Lundberg and Lee, 2017) decomposes each prediction additively, $f(x) = \phi_0 + \sum \phi_j(x)$, here in log-odds of dropout; the mean absolute SHAP value ranks the global drivers. LIME (Ribeiro et al., 2016) fits a sparse local surrogate around one prediction. Gender and Marital status are not model inputs, so they cannot appear in either explanation.

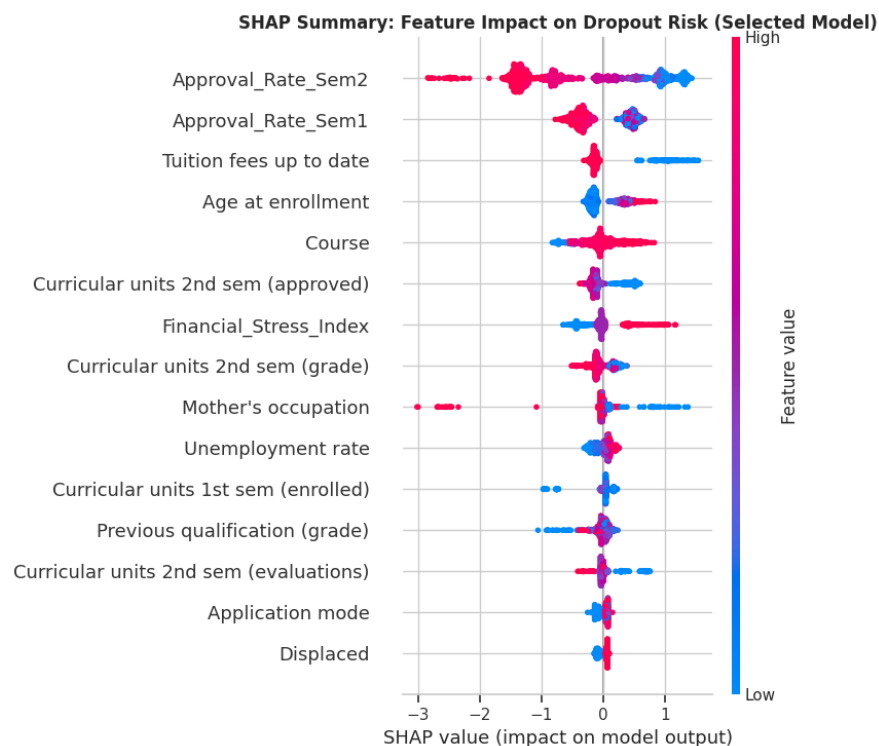


Figure 4. SHAP summary (beeswarm) for the selected model. Each dot is a hold-out student; colour shows the feature value.

Table 6. Ten largest global drivers (mean |SHAP|, log-odds of dropout)

Rank	Feature	Mean SHAP	Rank	Feature	Mean SHAP
1	Approval_Rate_Sem2	1.052	6	Curricular units 2nd sem (approved)	0.203
2	Approval_Rate_Sem1	0.409	7	Financial_Stress_Index	0.199
3	Tuition fees up to date	0.262	8	Curricular units 2nd sem (grade)	0.144
4	Age at enrollment	0.238	9	Mother's occupation	0.123
5	Course	0.230	10	Unemployment rate	0.108

- Higher approval rates and up-to-date tuition lower predicted risk. Three of the top ten drivers (both approval rates and Financial_Stress_Index) are features engineered in Section 3, and three are Semester 2 variables, which explains why waiting until the end of Semester 2 improves detection. Grade_Trend is not among the top ten.
- For one correctly flagged student (index 3744, predicted risk 97.1%), LIME shows that Approval_Rate_Sem2 at or below 0.40 contributes most (+0.251), Financial_Stress_Index at or below 1 reduces risk (-0.156), and Nacionality at or below 1 adds +0.107 (Figure 5). Two of LIME's top three features also rank in the SHAP top ten, so the two methods only partly agree and local explanations should be read with care.

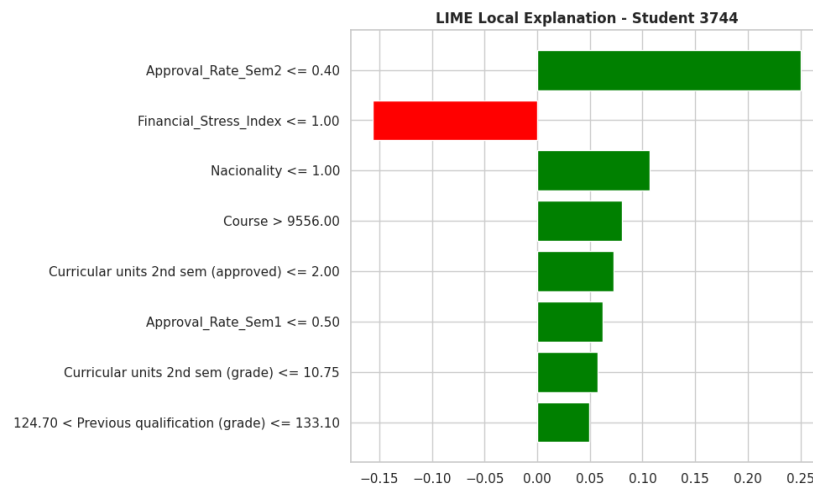


Figure 5. LIME local explanation for one correctly flagged dropout (hold-out student 3744).

7. Bias and Fairness Analysis

7.1 Audit design

The audit covers five attributes: gender, marital status (single or not), scholarship status (a socio-economic proxy), debtor status and age group (20 or under, 21 to 25, over 25). Gender and Marital status come from the separate audit table because they are not model inputs. Groups with fewer than 30 students are excluded from ratios. For each group the audit computes selection rate, recall (true-positive rate), false-positive rate (FPR) and precision, and for each attribute:

- **Disparate impact (flag-rate ratio)** = lowest group selection rate / highest group selection rate; values below 0.80 are flagged (a screening heuristic borrowed from US employment practice, used here as a warning flag, not as a legal test).
- **Equal opportunity** = the gap between the highest and lowest group recall (Hardt et al., 2016); gaps above 0.10 are flagged.

- **Equalized odds** requires both the recall gap and the FPR gap to be small.

Recall parity is treated as the primary criterion: selection-rate gaps partly reflect real differences in base dropout rates between groups, whereas unequal recall means one group's at-risk students are missed more often, which is the harm that matters here.

7.2 Findings

Table 7. Per-group performance at threshold 0.50 (hold-out set)

Attribute	Group	N	Dropout rate	Selection rate	Recall	FPR	Precision
Gender	Female	564	0.245	0.342	0.906	0.160	0.648
Gender	Male	321	0.455	0.576	0.918	0.291	0.724
Marital status	Not single	110	0.527	0.627	0.948	0.269	0.797
Marital status	Single	775	0.292	0.399	0.903	0.191	0.660
Scholarship	No scholarship	654	0.390	0.520	0.925	0.261	0.694
Scholarship	Scholarship	231	0.126	0.165	0.793	0.074	0.605
Debtor	Debtor	90	0.578	0.756	0.981	0.447	0.750
Debtor	Not debtor	795	0.292	0.390	0.897	0.181	0.671
Age group	21-25	157	0.357	0.522	0.946	0.287	0.646
Age group	20 or under	513	0.199	0.288	0.804	0.161	0.554
Age group	Over 25	215	0.586	0.688	0.984	0.270	0.838

Table 8. Fairness metrics per attribute

Attribute	Flag-rate ratio	Recall gap	FPR gap	Equalized-odds gap	Flags
Gender	0.594	0.012	0.132	0.132	flag-rate ratio
Marital status	0.636	0.046	0.078	0.078	flag-rate ratio
Scholarship holder	0.316	0.132	0.186	0.186	flag-rate ratio; recall gap
Debtor	0.516	0.084	0.266	0.266	flag-rate ratio
Age group	0.419	0.180	0.127	0.180	flag-rate ratio; recall gap

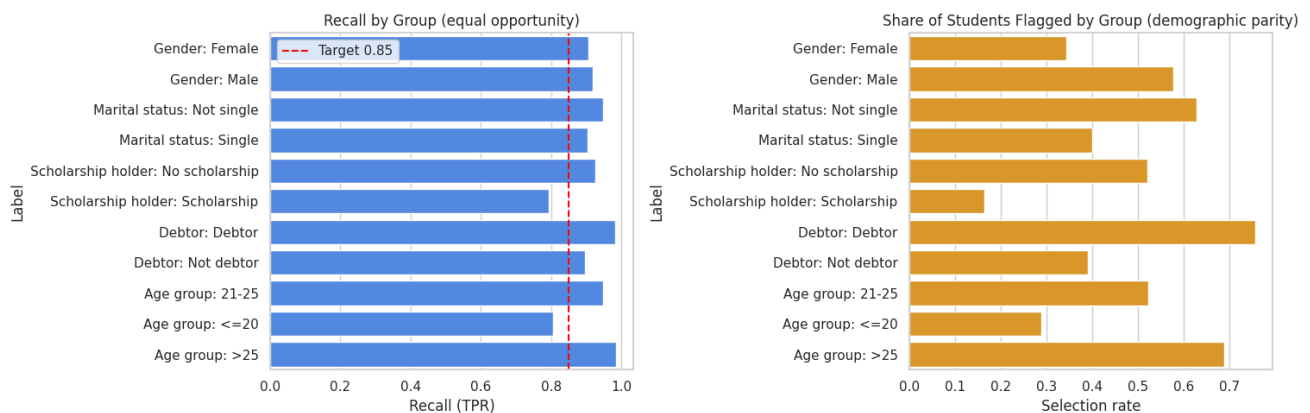


Figure 6. Recall by group (left, equal opportunity) and share of students flagged by group (right, demographic parity).

- **Flag-rate ratios** fall below 0.80 for all five attributes, but this largely reflects different base dropout rates: for example scholarship holders drop out at 12.6% against 39.0% for others, and students aged 20 or

under at 19.9% against 58.6% for those over 25.

- **Recall gaps** exceed 0.10 for two attributes. Students aged 20 or under (recall 0.804 against 0.984 for those over 25) and scholarship holders (0.793 against 0.925) have more of their at-risk students missed. Gender, marital status and debtor status are within the 0.10 limit.
- **Excluding sensitive attributes:** a counterfactual model that does receive Gender and Marital status has the same ROC-AUC (0.937), slightly lower recall (0.905) and a larger gender recall gap (0.041 against 0.012). Excluding them therefore cost nothing in accuracy, although other features can still act as proxies, which is why outcomes are audited by group rather than assumed fair.

7.3 Mitigation, implemented and evaluated

Two mitigations were implemented for the attribute with the largest recall gap, age group, and scored on the untouched hold-out set against the baseline. **Reweighting** (in-processing; Kamiran and Calders, 2012) gives each training row the weight $w(g, y) = P(g) P(y) / P(g, y)$, which makes group and outcome statistically independent in the weighted training set, and the model is refitted. **Group-specific thresholds** (post-processing) choose one decision threshold per age group on out-of-fold training predictions so that each group's recall matches the overall level of 0.880 (thresholds: 20 or under 0.30; 21 to 25 0.53; over 25 0.79); the test set is never consulted when choosing them. The decision rule was fixed in advance: adopt a mitigation only if it narrows the recall gap while overall recall stays at or above 0.85.

Table 9. Mitigation comparison on the hold-out set (age group)

Variant	Recall	Precision	F2	ROC-AUC	Flag-rate ratio	Recall gap	FPR gap
Baseline (selected model)	0.912	0.685	0.855	0.937	0.419	0.180	0.127
Reweighting (in-processing)	0.901	0.642	0.834	0.935	0.552	0.139	0.127
Group thresholds (adopted)	0.898	0.614	0.822	0.937	0.741	0.074	0.214

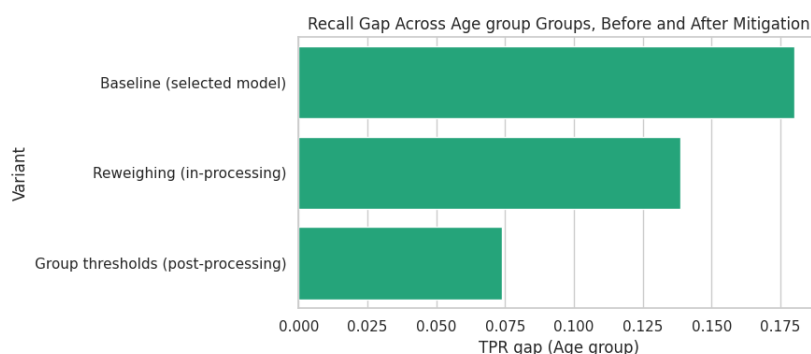


Figure 7. Recall gap across age groups before and after mitigation.

The rule adopts group thresholds, which cut the age-group recall gap from 0.180 to 0.074 and raise the flag-rate ratio from 0.419 to 0.741, at the cost of recall (0.912 to 0.898) and precision (0.685 to 0.614). **Caveats:** the false-positive gap widens from 0.127 to 0.214, so the equalized-odds gap is not improved (0.180 to 0.214); age is used at decision time, which should be confirmed as acceptable under institutional policy; the scholarship gap (0.132) was not targeted and remains; and nationality and international-student status are still model inputs that have not yet been audited.

8. Business Implications

The business briefing translates the measured results into decision terms. At the adopted operating point (recall 0.898, precision 0.614), a cohort the size of the dataset (4,424 students, 1,421 dropouts) would see about 1,276 dropouts caught and about 802 students flagged who would have stayed, so advisors would contact about 2,078 students, roughly 47% of the cohort. **Advisory capacity, not model accuracy, is therefore the practical constraint.**

An illustrative return calculation uses planning assumptions that come from the project brief, not from the dataset: tuition value of \$24,000 per retained student, 25% of caught dropouts retained through intervention, and programme cost scaling at about \$123 per flagged student. On these assumptions about 319 students are retained, preserving about \$7.66 million of multi-year tuition against a programme cost of about \$256 thousand (net return 28.9 times cost). Break-even requires retaining only 0.8% of caught dropouts. Because the inputs are assumptions, the table below shows the sensitivity of the return to them; a pilot should replace them with measured values.

Table 10. Illustrative net return per \$1 of programme cost (planning assumptions, not dataset facts)

Tuition value per student	5% retained	10% retained	25% retained (base case)
\$24,000 (base case)	5.0x	11.0x	28.9x
\$12,000	2.0x	5.0x	14.0x
\$6,000	0.5x	2.0x	6.5x

9. Limitations, Recommendations and Future Work

9.1 Limitations

- **Class imbalance:** 32.1% dropouts; class weighting raises recall at the cost of precision (extra false alarms).
- **Label censoring:** students still Enrolled are labelled Retained although their final outcome is unknown.
- **Generalisability:** one institution, one country, one cohort period; performance must be re-validated before use elsewhere.
- **Proxies and unaudited inputs:** excluding Gender and Marital status does not remove correlated information, and nationality and international status remain unaudited model inputs.
- **Small groups:** subgroups below 30 students are excluded from fairness ratios, so their treatment cannot be certified from this data.
- **Unobserved causes:** illness, family and work changes are invisible to administrative records.
- **Model selection margin:** the top models are close in cross-validated F2, so the choice among them is partly a judgement call.
- **Mitigation trade-offs:** age-specific thresholds widen the false-positive gap and raise a policy question; the scholarship gap is not addressed.

9.2 Recommendations and future work

- **Operate** the end-of-Semester-2 scoring as advisor decision support, not automated action; every flag triggers a human check-in, matched to the dominant risk signal (financial-aid review, peer tutoring and advising, or mature-student support).
- **Monitor** recall, flag rate and per-group recall gap each cycle, and retrain when drift or a gap above 0.10 appears.

- **Audit next:** nationality and international status, and the scholarship recall gap.
- **Validate** on a new cohort or institution, and **measure** real retention lift and advisor workload in a pilot to replace the planning assumptions.
- **Extend** with probability calibration, threshold review under capacity constraints and, if data allows, a time-to-dropout formulation.

10. Reproducibility and Repository

The complete analysis is a single notebook that runs top to bottom (fixed seeds, stratified split, preprocessing fitted on training data only, selection on training cross-validation, one scoring of the test set). Step 4 writes every tuned model, the comparison table and a configuration file; the Step 6 cell uses nbconvert to generate an HTML report, interactive slides, a LaTeX file and a PDF directly from the notebook. The public GitHub repository is organised like an open-source project:

Table 11. Repository contents

Path	Contents
notebooks/	The executed analysis notebook (Steps 1 to 7)
src/	Helper modules extracted from the notebook: feature engineering and fairness audit functions
data/	Documentation for the dataset, which the notebook downloads automatically
models/	Tuned models (.joblib), model_comparison.csv and best_model_config.json
reports/	This final report, the technical and business presentations, the nbconvert exports
tests/	Tests for the helper modules
README.md, requirements.txt	Overview, results, quickstart, limitations, citation, and the dependency list

Repository: https://github.com/theboneidentity/PILLAR_5_CAPSTONE_PROJECT

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